The Prisoner's Dilemma

From Game Theory to Spatial ABM and Reinforcement Learning

From Conway to LangGraph — Bridge Lecture (Weeks 5→ 6)

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I. Setting Up the Game

Two suspects, two choices, one dilemma

THE STORY

Two suspects are interrogated in separate rooms. Neither can communicate. Each must choose independently:

Cooperate (C) — stay silent

Defect (D) — betray the other

The payoff matrix shows the outcome for the row player:

	B Cooperates	B Defects
A Cooperates	R , R	S , T
A Defects	T , S	P , P

The Four Payoff Parameters

$T > R > P > S$ and $2R > T + S$

$$T > R > P > S \quad \text{and} \quad 2R > T + S$$

T	R	P	S
Temptation	**Reward**	**Punishment**	**Sucker**
Defecting vs a cooperator. Best individual outcome. Allure of betrayal.	Mutual cooperation. Social optimum. What both want collectively.	Mutual defection. Nash equilibrium outcome. Individually rational, collectively bad.	Cooperating vs a defector. Worst individual outcome. Cost of naive generosity.

Board example: T=5, R=3, P=1, S=0 | Nowak-May spatial model: T=b>1, R=1, P=0, S=0

Nash Equilibrium

No player can benefit by unilaterally changing strategy

DEFINITION

$$(s_A^*, s_B^*) \text{ is Nash} \iff \pi_A(s_A^*, s_B^*) \geq \pi_A(s_A, s_B^*) \;\; \forall s_A \quad \text{and} \quad \pi_B(s_A^*, s_B^*) \geq \pi_B(s_A^*, s_B) \;\; \forall s_B$$

DOMINANT STRATEGY ARGUMENT (draw at board)

Fix A's strategy.

If B cooperates: A gets T=5 (defect) > R=3 (cooperate).

If B defects: A gets P=1 (defect) > S=0 (cooperate).

Conclusion: Defection dominates regardless of what B does. By symmetry, the same holds for B.

(D, D) is the unique Nash equilibrium — in fact a dominant strategy equilibrium

The Dilemma: (D,D) yields P=1 to both. Mutual cooperation (C,C) would yield R=3. Individual rationality leads to collective sub-optimality.

Pareto Optimum — Classifying All Four Outcomes

(s_A,s_B) is Pareto-optimal if no outcome makes one player strictly better without making the other worse

No feasible (s_A', s_B') exists with $\pi_A' \geq \pi_A$ and $\pi_B' \geq \pi_B$ and at least one strict

(C,C)

Payoff: (3, 3)

Nash: No

Pareto: Yes ✓

A can deviate to D → get 5. (D,C),(C,D),(D,D) all make at least one player worse.
→ No Pareto-dominating alternative.

(D,D)

Payoff: (1, 1)

Nash: Yes

Pareto: No ✗

Neither can improve unilaterally. (C,C) gives (3,3) — better for BOTH.
→ Only Pareto-suboptimal outcome.

(D,C)

Payoff: (5, 0)

Nash: No

Pareto: Yes ✓

B can deviate to D → get P=1>0. A already has maximum T=5. Improving B requires reducing A.

(C,D)

Payoff: (0, 5)

Nash: No

Pareto: Yes ✓

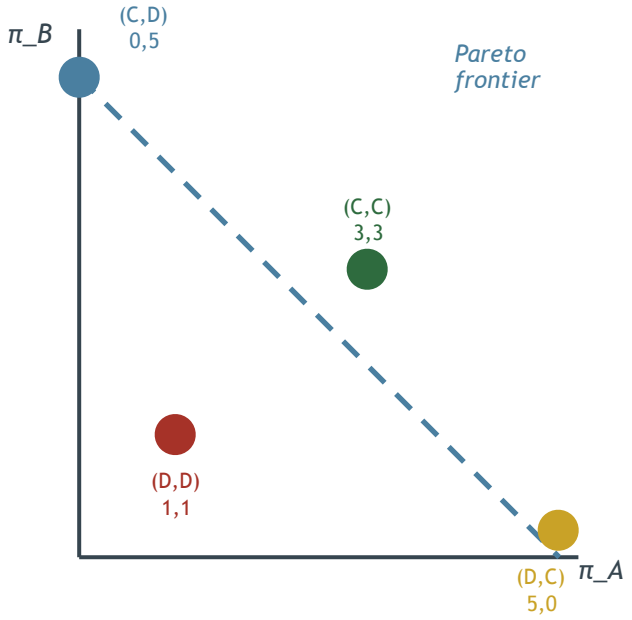
Symmetric to (D,C). Pareto-optimal, not Nash.

(D,D) is the ONLY Pareto-suboptimal outcome. Every other outcome Pareto-dominates it. Pareto frontier = {(C,C),(D,C),(C,D)}. Rational self-interest selects the uniquely worst social outcome.

The Pareto Frontier and the Free Energy Landscape

The social potential, the Nash trap, and the nucleation analogy

PAYOFF SPACE (π_A vs π_B)



PHYSICS — FREE ENERGY LANDSCAPE

(C,C) = global minimum

of social potential U = −(π_A + π_B). Total payoff 6.

(D,D) = Nash trap

Saddle point of social potential. Local minimum of each player's individual free energy.

Pareto frontier

The constraint surface in payoff space — set of efficient allocations.

Escape requires nucleation

Reaching (C,C) from (D,D) requires coordinated motion. Neither player can move unilaterally — like nucleation in a first-order phase transition.

Utilitarian social optimum (max sum of payoffs) = (C,C) with total 6. Condition 2R>T+S ensures (C,C) beats even the unequal outcomes (T+S=5 < 6).

II. The Iterated Prisoner's Dilemma

Repetition alone is not enough — conditional strategies are

DISCOUNTED TOTAL PAYOFF

$$G = \Sigma_{t=0}^{\infty} \delta^t \cdot \pi_t = \pi_0 + \delta \pi_1 + \delta^2 \pi_2 + \ldots \qquad \delta \in (0,1)$$

Two interpretations of δ: (1) game ends with prob (1−δ) after each round | (2) future payoffs discounted by δ per period. Formally equivalent.

Wrong intuition

If A plays C always: $G = R/(1-\delta)$.
If A defects always vs naive AllC: $G = T/(1-\delta) > R/(1-\delta)$.

Long horizon changes nothing — perpetual defection still beats unconditional cooperation.

Why one-shot logic persists

Each individual round is still a PD.
At any given round, defection yields a strictly higher payoff — exactly as in the one-shot game.

Myopic reasoning → defect every time.

What enables cooperation

Two ingredients required:

1. Memory — observe past outcomes.
2. Conditioning — future action depends on history.

Only then can a player credibly threaten punishment.

The correct calculation: compare cooperation path vs best deviation under a conditional strategy (Grim Trigger) — Section 2.2.

Grim Trigger — Full Derivation

The simplest conditional strategy that enforces cooperation

GRIM TRIGGER STRATEGY

Round 1: Play C.
Round t>1: Play C if opponent always played C.
Otherwise play D forever.

Step 1 — Mutual cooperation payoff:

$$V(\text{coop}) = R/(1-\delta)$$

Step 2 — Best deviation payoff:

Defect at t=0 → get T. GT triggers → both play D forever (P each round).

$$V(\text{deviate}) = T + \delta P/(1-\delta)$$

Stationarity: deviating later gives the same value — can't do better.

STEP 3 — INCENTIVE COMPATIBILITY

Cooperation is self-enforcing iff $V(\text{coop}) \geq V(\text{deviate})$:

$$R/(1-\delta) \geq T + \delta P/(1-\delta)$$

$$R \geq T(1-\delta) + \delta P \quad [\times (1-\delta)]$$

$$R \geq T - \delta T + \delta P$$

$$R - T \geq \delta(P - T)$$

$$(T - R) \leq \delta(T - P) \quad [\div (T-P) > 0]$$

$$\delta \geq (T - R) / (T - P)$$

Interpreting the Cooperation Threshold

$\delta^* = (T-R)/(T-P)$ — with $T=5$, $R=3$, $P=1$ → $\delta^* = 0.5$

$$\delta^* = (T - R) / (T - P) = (5-3)/(5-1) = 2/4 = 0.5$$

$\delta = 0$ — fully myopic

Agents care only about the current round.
$Q^*(s,D) > Q^*(s,C)$ everywhere.
One-shot Nash equilibrium re-emerges automatically.
Cooperation impossible.

$\delta = \delta^*$ — threshold

Agent weights tomorrow at half of today.
Cooperation and defection exactly indifferent.
GT is a Nash equilibrium — barely.
Small increase in δ tips the balance.

$\delta \to 1$ — patient

Future rounds matter as much as today.
Cooperation easy to sustain.
Almost any T,R,P,S satisfies $\delta^* < 1$.
Folk Theorem very permissive.

Physics analogy — activation energy: $(T-R)$ = energy barrier. $\delta \cdot (T-P)$ = long-run penalty weighted by future horizon. Low δ = high effective temperature: thermal fluctuations (myopia) overcome the barrier and the system falls into the defection ground state.

2.3 Strategy Survey

AllD · AllC · Tit-for-Tat and their match-ups

Always-D (AllD)

- Dominant in one-shot game.
- vs AllC: $T/(1-\delta)$. Fully exploits.
- vs AllD: $P/(1-\delta)$. Mutual trap.
- vs GT/TfT: $T+\delta P/(1-\delta)$. Wins if $\delta < \delta^*$.
- Evol. weakness: in all-D population, cooperator cluster can invade via spatial structure.

Always-C (AllC)

- vs AllC: $R/(1-\delta)$. Social optimum.
- vs AllD: $S/(1-\delta)$. Fully exploited.
- Not ESS: single AllD mutant does strictly better.
- Shows that unconditional cooperation is not the goal — conditional cooperation is.

Tit-for-Tat (TfT)

Start C; copy opponent's last move.

- Nice — never defects first.
- Retaliatory — punishes immediately.
- Forgiving — one defection → one punishment.
- Clear — opponents quickly understand it.

⚠ **Echo effects:** A misimplemented move triggers mutual punishment spirals. Two TfT players can lock into permanent mutual defection from a single error.

GTFT · WSLS · Strategy Comparison Table

Noise-robust strategies and the full summary

Generous Tit-for-Tat (GTFT)

$$p_forgive = \min(\ 1 - T/R,\ \ (R-S)/(T-S)\ )$$

Like TfT but forgives defections with prob p. Breaks echo spirals. GTFT displaced TfT in noisy populations (Nowak & Sigmund 1992). Optimal p balances noise-robustness vs exploitability.

Win-Stay Lose-Shift (WSLS / Pavlov)

Repeat if $\pi_t \in \{R,T\}$ (good). Switch if $\pi_t \in \{P,S\}$ (bad).

Self-corrects (D,D): both get P (bad) → both switch to C. Recovers from noise in 2 rounds. (C,C) is a fixed-point attractor; (D,D) is a repeller.

Strategy	vs AllC	vs AllD	vs Self	Noise?	Fixes (D,D)?
AllD	T/(1−δ) ✓	P/(1−δ)	P/(1−δ)	—	No
AllC	R/(1−δ)	S/(1−δ) ✗	R/(1−δ)	—	No
GT	R/(1−δ)	S+δP/(1−δ)	R/(1−δ)	No ✗	No
TfT	R/(1−δ)	S+δP/(1−δ)	R/(1−δ)	No ✗	No
GTFT	R/(1−δ)	Better AllC	R/(1−δ)	Yes ✓	Slow
WSLS	R/(1−δ)	~(S+T)/2	R/(1−δ)	Yes ✓	Yes ✓

III. Spatial PD — Nowak & May (1992)

Square lattice · Moore neighbourhood · Synchronous update · Copy-the-Best

MODEL SETUP

Lattice :	L×L square, periodic boundaries (torus)
Strategy :	$s_i \in \{C,D\}$ at each site
Neighbourhood :	Moore: 8 neighbours + self = 9 interactions
Payoff :	$\Pi_i = \Sigma_{j\in\mathcal{N}(i)\cup\{i\}} \pi(s_i, s_j)$
Update :	Copy-the-Best — synchronous
Parameter :	Single control: b > 1 (benefit of defection)

PAYOFF MATRIX + UPDATE RULE

C vs C → 1	C vs D → 0
D vs C → b	D vs D → 0

TWO-PHASE GENERATION

$$j^* = \mathrm{argmax}_{j\in\mathcal{N}(i)\cup\{i\}} \Pi_j(t)$$

$$s_i(t+1) = s_{j^*}(t)$$

Phase 1: all agents accumulate payoffs using the time-t snapshot. No updates yet.
Phase 2: all agents simultaneously adopt the strategy of the highest-payoff agent in their neighbourhood.

Synchronous → period-2 oscillations in Phase I.

The Three Phases of the Spatial PD

MF: extinction for all b>1 · Spatial: three distinct regimes

Phase I

$1 < b < 1.8$

Stable Coexistence

Cooperators and defectors coexist indefinitely in dynamically evolving, spatially structured patterns.

Asymptotic $f_C > 0$, depends on b.

Cluster interiors shield cooperators from exploitation.

MF: extinction
Spatial: coexistence
→ MF qualitatively WRONG

Phase II

$1.8 \lesssim b < 2$

Metastable Cooperators

IC-dependent outcome.

Defectors usually dominate from random ICs but cooperator clusters survive as metastable islands.

Phase boundary not sharp — crossover sensitive to L.

MF: extinction
Spatial: IC-dependent
→ Spinodal region

Phase III

$b \geq 2$

Defectors Dominate

PD condition fails:
$T+S = b \geq 2 = 2R$

Even spatial structure cannot prevent cooperator extinction.

MF and spatial agree qualitatively.

MF: extinction
Spatial: extinction
→ MF is correct

Key: for $1 < b < 1.8$, spatial structure creates a cooperation phase with NO analogue in the well-mixed theory. MF misses an entire phase.

The Mechanism: Spatial Reciprocity

Cluster payoff geometry — the decisive comparison

Interior cooperator

9 interactions, all C.
Payoff = 8×1 + 1×1 = 9

Completely shielded from defectors.
The protected core of every cluster.

Edge cooperator

Has k_D defector neighbours.
Payoff = 9 − k_D

Partially exploited but still earns positive payoff from cooperating neighbours.

Edge defector

Has k_C cooperator neighbours.
Payoff = $k_C \times b$

If k_C is small, $k_C \cdot b$ may be LESS than the cooperator edge payoff $9-k_D$.

Interior defector

Surrounded by defectors.
Payoff = 0

Defectors cannot exploit each other.
The key payoff collapse.

Decisive comparison: Interior cooperator earns 9. Interior defector earns 0. The cluster wins the *internal competition* once the boundary stabilises — like positive surface tension shrinking defector domains.

Domain Wall Dynamics · Network Topology

Ising analogy and the role of graph structure

ISING ANALOGY — DOMAIN WALLS

Cooperators → spin +1 Defectors → spin −1
Cluster boundary = domain wall

Phase I (1<b<1.8)

Positive surface tension. Domain wall shrinks defector regions. Cooperator clusters thermodynamically stable.

Phase II (1.8<b<2)

Spinodal region. Metastable states — like region between binodal and spinodal in liquid-gas transition.

Phase III (b≥2)

Reversed surface tension. Defector regions grow. Cooperator clusters vanish.

⚠ Nowak-May is NOT an equilibrium model. No Hamiltonian, no detailed balance. "Phases" are dynamical attractors, not thermodynamic phases. Fermi rule does restore detailed balance.

NETWORK TOPOLOGY

Regular lattice

High clustering → spatial reciprocity → high f_C.

Small-world

Long-range shortcuts can destabilise clusters. Mixed effect on f_C.

Random graph

Low clustering → well-mixed limit → cooperation collapses.

Scale-free (BA)

Hubs pivotal: cooperating hubs propagate cooperation. Santos & Pacheco (2005): degree heterogeneity acts as spatial reciprocity proxy.

Percolation analogy: cooperation threshold ↔ percolation threshold. Universality class depends on network topology.

IV. Update Rules

Copy-the-Best (T=0) vs Fermi Rule (T=κ)

Copy-the-Best (Imitate-the-Best)

$$j^* = \mathrm{argmax}_{j \in \mathcal{N}(i) \cup \{i\}} \Pi_j$$
$$s_i(t+1) = s_{j^*}(t)$$

- Deterministic — no randomness.
- Effective temperature = 0.
- Can be trapped in local optima.
- Synchronous → period-2 oscillations.
- RL: greedy policy (ε=0, κ→0).

Fermi Rule (Szabó & Tőke 1998)

$$P(s_i \leftarrow s_j) = 1 / [1 + \exp(-(\Pi_j - \Pi_i)/\kappa)]$$

- κ→0: step function → Copy-the-Best (T=0).
- κ→∞: P→1/2 → random copying (T=∞).
- Finite κ: prevents premature convergence.
- Restores detailed balance → thermodynamic model.
- RL: Boltzmann / softmax policy.

Property	Copy-the-Best	Fermi Rule
Noise	None (deterministic)	Tunable via κ
Physics analogy	Zero-temperature spin dynamics	Finite-T Glauber / Metropolis
RL analogy	Greedy policy (ε=0)	Boltzmann / softmax policy
Exploration	None — exploits only	Yes — explores strategy space

V. Mean-Field Theory

The physicist's Hello World — and why it fails

REPLICATOR EQUATION DERIVATION

Well-mixed: every agent interacts with every other with equal probability.

$$\langle \Pi_C \rangle = f_C \cdot 1 + (1-f_C) \cdot 0 = f_C$$

$$\langle \Pi_D \rangle = f_C \cdot b + (1-f_C) \cdot 0 = b \cdot f_C$$

Since $b > 1$: $\langle \Pi_D \rangle > \langle \Pi_C \rangle$ for any $f_C > 0$.

$$df_C/dt = f_C^2(1-f_C)(1-b) < 0 \quad \text{for } b > 1$$

f_C decreases monotonically to zero for any $b > 1$, any initial $f_C \in (0,1)$.

FIXED POINTS

$f_C^* = 0$ (all defect)

Stable attractor for $b>1$

$f_C^* = 1$ (all cooperate)

Unstable for $b>1$

$f_C^* \in (0,1)$

Only at $b=1$ exactly

MF verdict:
Cooperators go extinct for ANY $b>1$.
Only stable fixed point: $f_C^* = 0$.

Spatial result: coexistence for $1<b<1.8$.

MF is qualitatively wrong.

Pair approximation (Bethe analogue) tracks C-C, C-D, D-D pair densities → predicts finite cooperation threshold. Exact threshold requires RG methods or Monte Carlo.

VI. Bridge to Reinforcement Learning

Every PD concept maps exactly onto the MDP framework

Game Theory / ABM	Reinforcement Learning / MDP
Strategy $s_i \in \{C,D\}$	Action $a_t \in \{C,D\}$
Payoff matrix $\pi(s_i, s_j)$	Reward function $R(s,a,s')$
Opponent's last move	State $s_t \in \mathcal{S}$
4 states (CC, CD, DC, DD)	State space $\lvert\mathcal{S}\rvert = 4$
Discount factor δ	Discount factor γ
Total payoff $\Sigma\, \delta^t\, \pi_t$	Return $G_t = \Sigma\, \gamma^k\, r_{t+k}$
Fermi κ-noise	Boltzmann / softmax policy
Copy-the-Best	Greedy policy ($\varepsilon=0$, $\kappa \to 0$)
Replicator dynamics	Policy gradient ascent
MF fixed point f_C^*	Value function fixed point V^*
Evolutionary stable strategy	Nash equilibrium policy

The PD as an MDP · The Deep Connection

Q-learning solves the same equation as Grim Trigger

EXPLICIT MDP

$\mathcal{S} = \{(C,C),(C,D),(D,C),(D,D)\}$ $\mathcal{A} = \{C,D\}$

Q-update:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

$\gamma = 0$	Myopic. Learns AllD. Re-discovers one-shot Nash.
$\gamma = \gamma^*$	Threshold. Cooperation and defection exactly indifferent.
$\gamma > \gamma^*$	Far-sighted. Learns cooperation.

≡

THE DEEP CONNECTION

$$\text{GT:} \quad \delta \geq (T-R)/(T-P)$$

$$\text{RL:} \quad \gamma \geq (T-R)/(T-P) = \gamma^*$$

The GT derivation was implicitly solving the Bellman optimality equation for the PD all along.

RL makes this explicit, algorithmic, and extensible to cases with no closed-form solution.

$\gamma^* = \delta^*$ — not merely an analogy, but the same inequality.

Multi-Agent RL — why it's hard:

Non-stationarity (opponent's policy changes) · Equilibrium selection (which Nash?) · Spin-glass landscape (exponential local minima, PPAD-complete) · Credit assignment in networks